Homework 4 EP 471
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1. Parabolic: Heat Transport in a Beam

Assumptions: Assume 1D heat conduction (negligible radial heat loss), constant thermal conductivity (k), and that the ambient temperature (T_∞) remains constant.

A. This problem models the physical process of heat transport along a one-dimensional beam using the following partial differential equation:

$$\frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2}$$

To solve this, we discretize time (index k) and space (index i). For the nodes inside the beam, the finite differences approximation can be used:

FTCS: Calculate the future time step directly from the current one.

$T_i^{k+1} = T_i^k + r(T_{i-1}^k - 2T_i^k + T_{i+1}^k)$

BTCS: Solve a system of equations for all nodes simultaneously at the future time step.

$-rT_{i-1}^{k+1} + (1+2r)T_i^{k+1} - rT_{i+1}^{k+1} = T_i^k$

$r = \frac{k\Delta t}{\Delta x^2}$ is the ratio that dictates the relationship between the spatial and time steps.

Since this problem dictates heat fluxes at the ends of the beam rather than fixed temperatures, A ghost node is placed just outside the physical beam to approximate the spatial derivative.

- Left Boundary ($i = 0$): The input temperature gradient is $\frac{\partial T}{\partial x} = -200\ \mathrm{K/m}$. A central difference is $\frac{T_1 - T_{-1}}{2\Delta x} = q_0$, making the imaginary ghost node $T_{-1} = T_1 - 2\Delta x q_0$. Plugging this into the FTCS formula yields the left boundary update:
  $T_0^{k+1} = (1-2r)T_0^k + 2rT_1^k - 2r\Delta x q_0$
- Right Boundary ($i = N$): With an output temperature gradient of $\frac{\partial T}{\partial x} = 200\ \mathrm{K/m}$, the ghost node $T_{N+1} = T_{N-1} + 2\Delta x q_L$ gives the right boundary update:
  $T_N^{k+1} = 2rT_{N-1}^k + (1-2r)T_N^k + 2r\Delta x q_L$

B. While BTCS is unconditionally stable, FTCS is conditionally stable. If the timestep is too large relative to your spatial step, the numerical errors compound, and the simulation will blow up. The stability criterion requires $r \leq 0.5$. We can rearrange this to find the maximum stable timestep:

$$\Delta t_{max} = \frac{\Delta x^2}{2k}$$

Using the parameters from the assignment:

- $\Delta x = 0.5\ \mathrm{m}$
- $k = 0.1\ \mathrm{m^2/s}$

$$\Delta t_{max} = \frac{0.5^2}{2(0.1)} = \frac{0.25}{0.2} = 1.25\ \mathrm{s}$$

Due to the constraints of the assignment, the dt used is:

$$\Delta t = \frac{1}{2}\Delta t_{max} = \frac{1}{2}(1.25\ \text{s}) = 0.625\ \text{s}$$

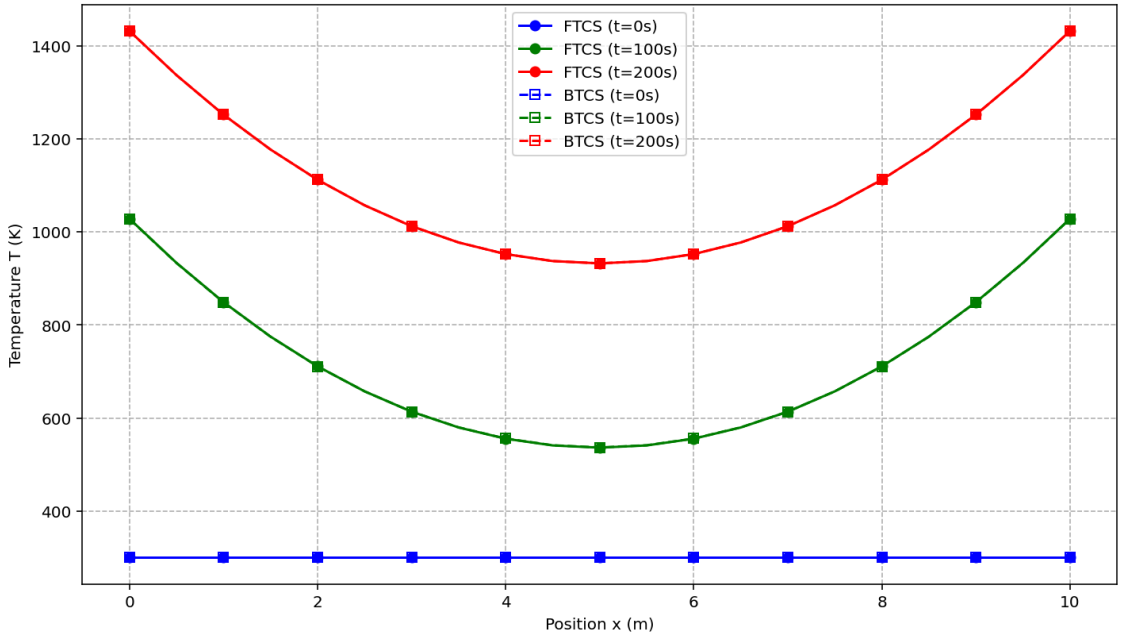


Figure 1: Temperature evolution in the beam at $t = 0, 100,$ and 200 s comparing FTCS and BTCS methods.

Both solving methods yield stable and accurate results. These match the expected results as heat is flowing into the beam at both ends at the same rate, and due to the lack of convection, a symmetric profile forms with a valley at the center, which is equidistant from the sources of heat. As time progresses, it is observed that the overall temperature of the bar increases, but the shape of the valley remains the same, the point-to-point slopes at each position does no tseem to vary between 100s and 200s.

C. Adding the convection term modifies the right-hand side of the heat equation. The continuous PDE is:

$$\frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2} + h(T_\infty - T)$$

Using the Backward Euler method in time and a centered difference in space, the implicit discretization is:

$$\frac{T_i^{k+1} - T_i^k}{\Delta t} = k\frac{T_{i-1}^{k+1} - 2T_i^{k+1} + T_{i+1}^{k+1}}{\Delta x^2} + h(T_\infty - T_i^{k+1})$$

Letting $r = \dfrac{k\Delta t}{\Delta x^2}$, move all future terms k+1 to the left side and group them:

$$-rT_{i-1}^{k+1} + (1 + 2r + h\Delta t)T_i^{k+1} - rT_{i+1}^{k+1} = T_i^k + h\Delta t T_\infty$$

For the boundary nodes, apply the Neumann ghost node substitutions:

$T_{-1}^{k+1} = T_1^{k+1} - 2\Delta x q_0$ and $T_{N+1}^{k+1} = T_{N-1}^{k+1} + 2\Delta x q_L$

- Left Boundary (x=0):
  $(1 + 2r + h\Delta t)T_0^{k+1} - 2rT_1^{k+1} = T_0^k + h\Delta t T_\infty - 2r\Delta x q_0$
- Right Boundary (x=L):
  $-2rT_{N-1}^{k+1} + (1 + 2r + h\Delta t)T_N^{k+1} = T_N^k + h\Delta t T_\infty + 2r\Delta x q_L$

By running and plotting, the 200s profile from the previous plot appears at the top, and the changes due to convection at the bottom. Convection from room temperature greatly reduces the temperature of the heated bar and widens the profile's valley.

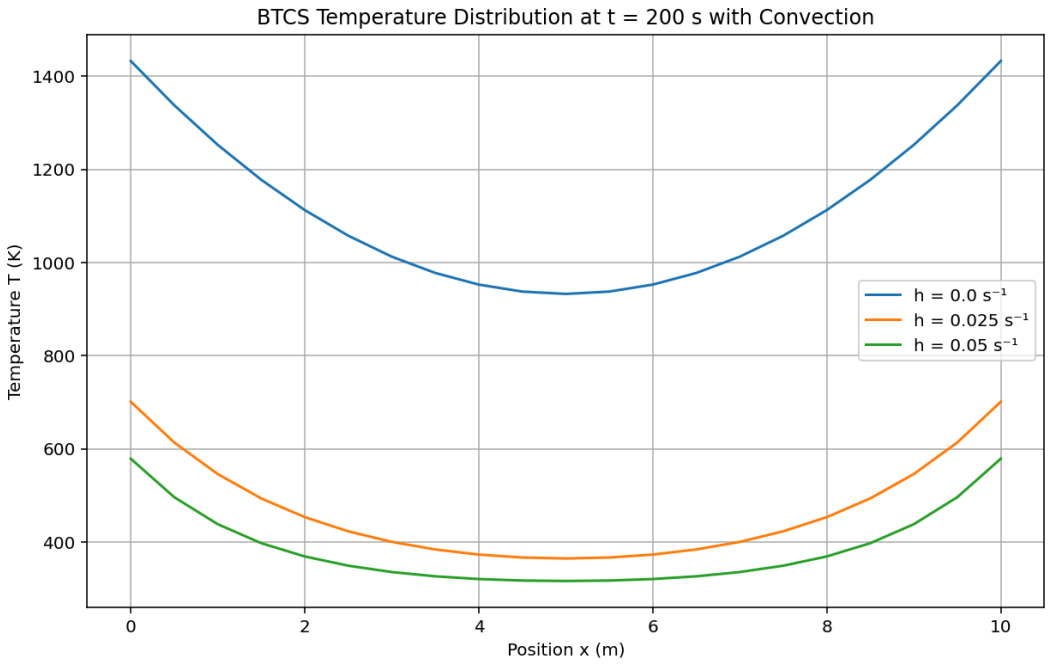


Figure 2: BTCS temperature distribution at $t = 200$ s for varying convection coefficients (h). When the convection term $h(T_\infty - T)$ is introduced, it acts as a stabilizing heat sink. As the beam's temperature rises above the ambient $T_\infty = 300$ K, the convection term begins to reject heat back into the surroundings.

2. Hyperbolic: Wave Equation

Assumptions: Assume a perfectly flexible string with constant tension and uniform linear density, leading to a constant wave speed (a).

A. Given the assumed solution format:

$u(x,t) = \sin(2\pi x)[A\sin(4\pi t) + B\cos(4\pi t)]$

Find the second temporal and spatial derivatives:

$u_{tt} = -16\pi^2 \sin(2\pi x)[A\sin(4\pi t) + B\cos(4\pi t)] = -16\pi^2 u$

$u_{xx} = -4\pi^2 \sin(2\pi x)[A\sin(4\pi t) + B\cos(4\pi t)] = -4\pi^2 u$

Substitute these into the wave equation $\dfrac{\partial^2 u}{\partial t^2} = a^2 \dfrac{\partial^2 u}{\partial x^2}$:

$-16\pi^2 u = a^2(-4\pi^2 u) \implies a^2 = 4 \implies a = 2$

Apply the initial displacement condition $u(x,0) = \sin(2\pi x)$:

$$\sin(2\pi x)[A(0) + B(1)] = \sin(2\pi x) \implies B = 1$$

Apply the initial velocity condition $\frac{\partial u}{\partial t}(x,0) = 4\pi \sin(2\pi x)$:

$$u_t(x,t) = \sin(2\pi x)[4\pi A \cos(4\pi t) - 4\pi B \sin(4\pi t)]$$

$$u_t(x,0) = \sin(2\pi x)[4\pi A] = 4\pi \sin(2\pi x) \implies A = 1$$

Therefore, $a = 2$, $A = 1$, and $B = 1$.

B. Using a standard first-order forward difference to initialize the first timestep relies solely on the initial velocity:

$$u_i^1 = u_i^0 + \Delta t \left(\frac{\partial u}{\partial t}\right)_i^0$$

This method introduces significant artificial diffusion. As plotted, the wave profile at $t = 0.3$ s visibly lags and differs in amplitude from the analytical solution because the first-order approximation poorly captures the initial acceleration of the wave.

C. By incorporating the Taylor series expansion from the PDE itself, we calculate the first step as:

$$u_i^1 = u_i^0 + \Delta t \left(\frac{\partial u}{\partial t}\right)_i^0 + \frac{C^2}{2}(u_{i+1}^0 - 2u_i^0 + u_{i-1}^0)$$

This second-order initialization correctly accounts for the spatial curvature at $t = 0$. The initial sine wave profile oscillates between the fixed boundaries (x=0, 1), and the standing wave pattern is maintained because both ends are fixed at zero displacement. At t=0.3s, the wave has undergone a partial oscillation from its initial state.

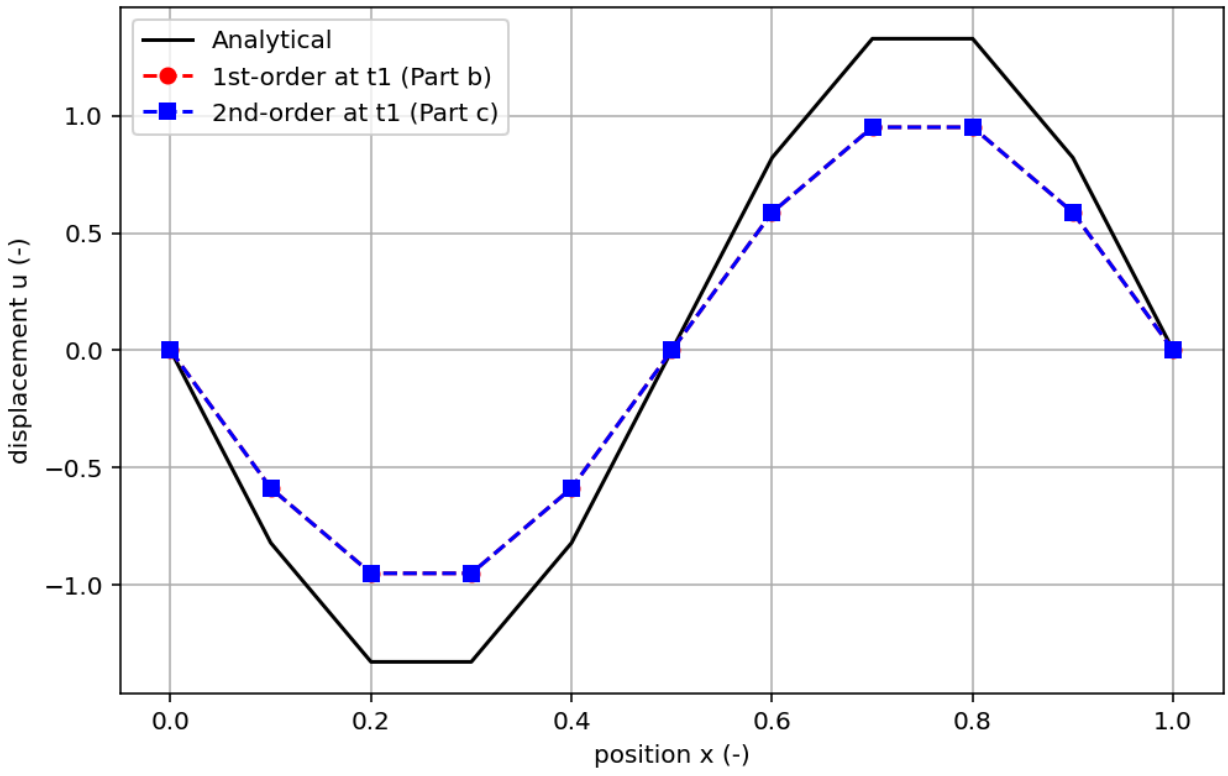
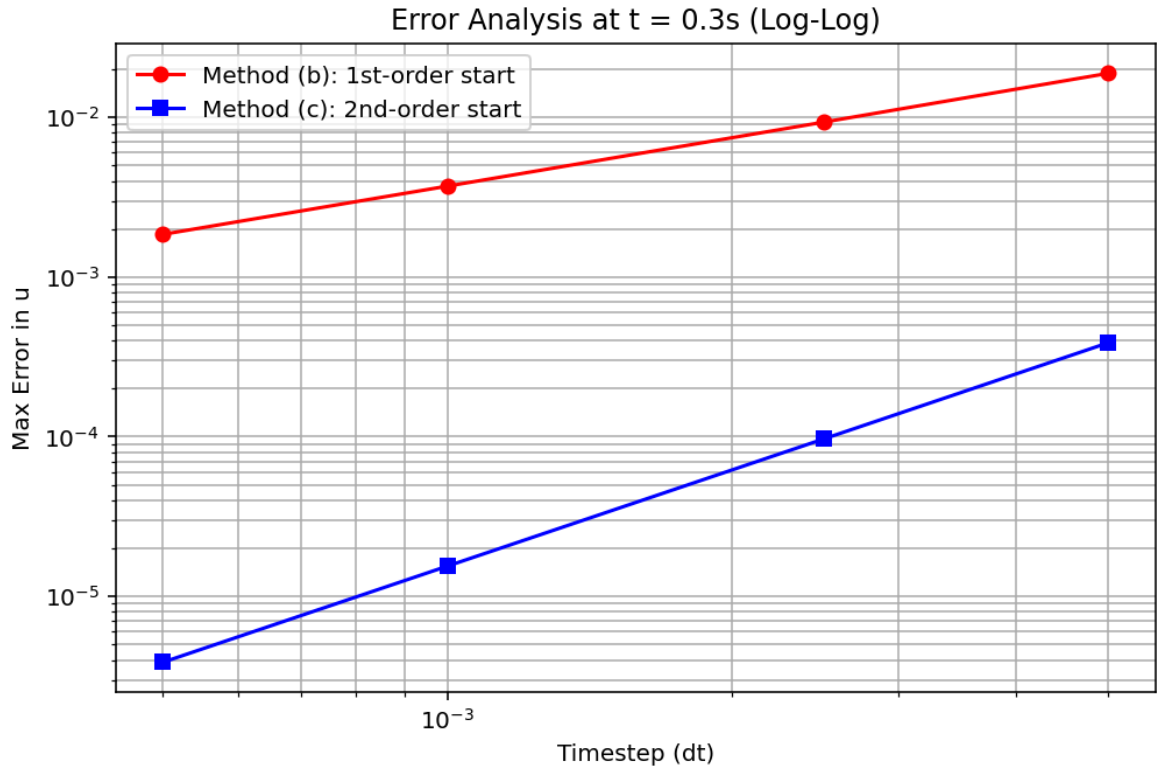


Figure 3: Wave displacement profile at t = 0.3 s comparing analytical results with 1st-order and 2nd-order numerical initializations.

Because $\Delta t = 0.05$ is relatively small, the $0.5\Delta t^2$ term is 0.00125. In a plot with a large spatial grid ($\Delta x = 0.1$), this tiny correction is visually indistinguishable until zoomed in or by looking at the error log-log plot.

D. Plotting the maximum error versus timestep on a log-log scale (while keeping the Courant number $C = \frac{a\Delta t}{\Delta x}$ constant at 1.0) reveals the order of accuracy of the schemes. It is hypothesized that the second-order temporal initialization will lead to a global error reduction rate of $\mathcal{O}(\Delta t^2)$, whereas the first-order start will limit the entire simulation to $\mathcal{O}(\Delta t)$ accuracy.



The slope of the error for the second-order corrected scheme is approximately 2, confirming it is $\mathcal{O}(\Delta t^2)$ accurate. The uncorrected first-order scheme yields a slope closer to 1, confirming its $\mathcal{O}(\Delta t)$ accuracy and demonstrating why the initial step is so critical in hyperbolic PDE simulations.